Master's Thesis Proposal

Working Title: Monetary Policy Transmission and Building Land Scarcity: A Panel Local Projections Analysis of Regional Austrian Housing Markets

Lea Schwabe (Student Number: 11908829)
University of Vienna – MSc Applied Economics

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Administrative Declaration:

- **Compulsory modules:** I hereby confirm that I have completed all three compulsory modules of the MSc Applied Economics (UA 066 913).
- **Potential supervisors:** My preferred supervisor for this thesis is **Assoz. Prof. Mag. Dr. Paul Pichler**. I am also open to supervision by another qualified faculty member as assigned by the program.

1 Introduction

In the period after the Global Financial Crisis (GFC), the European Central Bank (ECB) employed several unconventional monetary policy tools as the policy rate approached the zero lower bound (Benigno et al., 2023). In Austria, the period of ECB monetary easing coincided with an acceleration of residential real estate prices and a subsequent decoupling from underlying fundamentals (Belabed et al., 2022).

1.1 Observation

The ECB's unconventional monetary tools were primarily implemented through the Asset Purchase Programme (APP) and the Pandemic Emergency Purchase Programme (PEPP)(Benigno et al., 2023). After its introduction in 2015, the APP led to a persistent decline in euro area government bond yields, particularly targeting the longer end of the yield curve (Eser et al., 2019). While the APP was implemented to counter deflationary pressures and ease financing conditions (Benigno et al., 2023), the PEPP was a response to pandemic-induced financial stress (Herc, 2024). The announcement of the PEPP in March 2020 also created immediate and significant reductions in government bond yields at longer maturities (Herc, 2024).

Following the GFC, house prices showed moderate annual growth rates in the 2010s (4% in 2019) which was largely in line with income levels (Belabed et al., 2022). However, prices accelerated sharply from 2020 onward, while real economic activity remained weakened. In 2020, the annual growth rate of real estate prices reached 7% and consecutively exceeded the 10% mark (y-o-y) for eight quarters, peaking at 12.6% growth in Q4/2021 (Belabed et al., 2022).

Analogous to the price developments, the OeNB fundamentals indicator shows that by the end of 2021, residential real estate prices were up to 30% above the fundamentally justified price level in Austria (Belabed et al., 2022). This overvaluation was mainly driven by price increases which outpaced the development of income and rents, indicating a growing decoupling effect of market prices and real economic conditions (Schneider, 2013; Belabed et al., 2022).

The discontinuation of the PEPP in 2022 marked a significant point in the Austrian real estate market. With the beginning of the interest-rate tightening policies, residential real estate prices started to decline (q-o-q) at the end of 2022, and the annual growth rate decreased to 1% by the beginning of 2023 (Hildebrandt et al., 2023). However, the correction did not fully reverse affordability implications of the preceding boom phase. The divergence between real estate prices and household income during the period of monetary easing significantly constrained access to homeownership, particularly in urban areas. This resulted in elevated price-to-income ratios even after the correction (Mundt et al., 2024).

This indicates that housing affordability in Austria is a structural phenomenon and cannot be linked solely to phases of monetary policy (Matějková and Tichá, 2025). This national trend does not consider regional variation. To understand price dynamics following changes in the monetary environment, it is necessary to examine how regional supply variations moderate the transmission of national shocks.

1.2 Relevance

While the development of residential real estate prices is well documented at the national level, there is an increasing body of literature on heterogeneous effects across regional housing markets. Monetary policy shocks have different effects across sub-national regions, where housing and land supply elasticities affect the absorption dynamics. Aastveit and Anundsen (2018) empirically argue that expansionary monetary policy shocks indicate

significantly stronger and more persistent price effects in supply-inelastic regions, when an increase in demand cannot be accommodated through increased construction. Further, it has been shown that the housing supply channel exacerbates regional price and distributional effects, as supply-constrained regions experience larger price increases (Albuquerque, 2024).

Austrian evidence aligns closely with these theoretical implications. Reiner and Musil (2023) document that land prices during the housing boom experienced such regional divergence, with the most pronounced increases concentrating in previously expensive and densely populated regions. They conclude that land price growth is negatively related to household income growth, which again indicates a decoupling effect of prices from income, particularly in supply-constrained regions where demand shocks were primarily absorbed through prices (Reiner and Musil, 2023).

This decoupling effect has direct implications in the affordability debate. Especially during periods of increased price growth, urban housing markets in Austria experienced price dynamics that outpaced income growth, widening regional affordability differences (Mundt et al., 2024). Effectively this implies that on a national level the price increases did subside with the start of the interest-rate hiking cycle, but regional divergence was evident in land prices (Reiner and Musil, 2023).

These findings provide a framework that links aggregate monetary policy shocks to heterogeneous regional housing market outcomes in Austria. In regions where housing and land supply are relatively inelastic, monetary easing predominantly translates to price increases rather than supply accommodation (Aastveit and Anundsen, 2018; Reiner and Musil, 2023). The interaction between monetary policy, supply elasticities and regional housing markets highlights the relevance of a regional analysis when assessing effects of ECB policy in Austria. While this theoretical framework establishes the link of supply constraints to price sensitivity, the causal and dynamic impact of monetary policy shocks on regional Austrian housing markets remains empirically unquantified.

1.3 Research Gap

Previous research has documented the dynamics of national residential real estate prices as well as a regional divergence of land prices. Existing Austrian research has primarily focused on descriptive trends to map real estate price developments to monetary policy phases (Mundt and Wagner, 2017; Reiner and Musil, 2023). While these studies confirm that prices have developed heterogeneously across regions, the specific impact of monetary policy has not yet been isolated from other macroeconomic factors.

Consequently, there is little evidence of the underlying causal relationship between euro area monetary policy and local market outcomes. Furthermore, the dynamic response of regional price development has not yet been directly linked to monetary policies.

The housing supply channel, whereby supply constraints exacerbate effects of monetary shocks, has largely been analyzed on the national level or via international comparison (Aastveit and Anundsen, 2018). This creates a relevant background for further research in the Austrian housing market. By focusing on national aggregates, regional divergences may not be fully captured. The Austrian housing market exhibits regional heterogeneity via significant price differences across districts, that align with variations in building land reserves. Average house prices (per m^2) ranged from €1.150–1.350 in districts such as Waidhofen an der Thaya, Gmünd and Oberwart to over €6.300–8.700 in Innsbruck-Stadt, Salzburg (Stadt) and Kitzbühel (STATISTIK AUSTRIA, 2025c)[1]. Districts with the lowest building land reserves, notably Salzburg (Stadt) and Innsbruck-Stadt, lie within the upper tail of the

[1]These descriptive figures are based on STATISTIK AUSTRIA data, the empirical analysis relies on an alternative dataset as outlined in section 2.1 Data.

price distribution, whereas districts with higher reserves like Güssing and Jennersdorf exhibit significantly lower price levels (ÖROK Monitoring of Building Land Reserves, 2025).

Existing research has shown that regional differences do exist in the Austrian residential real estate market. Studying the interaction between aggregate monetary policy shocks and regional land scarcity will bridge the gap to a granular understanding of how structural affordability issues are related to euro area monetary policy, and if they are a consequence of the monetary easing period or a permanent distortion due to regional supply elasticities.

1.4 Research Question

The primary objective of this thesis is to examine the moderating role of regional building land elasticity in the transmission of ECB monetary policy shocks to residential real estate prices in Austrian districts. While evidence shows that phases of monetary easing have substantially driven Austrian real estate prices and led to a decoupling effect from underlying fundamentals, this thesis aims to analyze whether Austrian districts have observed systematic regional heterogeneity driven by structural supply constraints, motivating the primary research question:

1. To what extent do local supply constraints moderate the dynamic response of district-level real estate prices to European Central Bank monetary shocks in Austria?

The central argument is driven by the interaction of a national macroeconomic shock (euro area monetary policy) and local real estate market conditions. While monetary policy is uniform across Austria, prior research has shown that effects on housing markets are not. Regions with inelastic building land supply are expected to adjust primarily via price effects, leading to more severe and more persistent price responses (Aastveit and Anundsen, 2018; Albuquerque, 2024).

Accordingly, this thesis is guided by secondary research questions:

a. Does the magnitude of cumulative log-changes in house prices differ systematically between supply-elastic and supply-inelastic districts in response to an aggregate monetary policy shock?

b. Are there differences in the temporal response of price effects across regions over a three-year period?

c. Are price responses asymmetric across monetary policy phases, such that supply-inelastic districts experience smaller reversal effects in the transition to monetary tightening?

By specifically adopting this spatial and temporal perspective by focusing on district level heterogeneity and the dynamic adjustment over time via local projections, this thesis examines how residential property prices on the district level react to a common monetary shock. Local projections allow for a flexible framework within which time-varying responses can be interpreted as direct interaction with regional peculiarities such as building land elasticity (Jordà, 2005, 2023).

2 Methodology and Data

To estimate the dynamic impact of monetary policy shocks, this thesis utilizes the local projections framework developed by Jordà (2005). Local Projections are robust to misclassifications and allow for estimation of impulse response functions at specific time horizons.

The following model will be specified for each quarter $h = 1, \ldots, 12$:

$$y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \tau_{t,h} + \beta_h \left(\text{Shock}_t \times \text{Inelastic}_i\right) + \theta_h X_{i,t} + \varepsilon_{i,t+h}. \tag{1}$$

where:

- $\alpha_{i,h}$ (District fixed effects): Controls for time-invariant local characteristics.
- $\tau_{t,h}$ (Time fixed effects): Controls for national, aggregate developments and ensures that β_h identifies the extent to which the response of inelastic districts differs from the national average.
- $\theta_h X_{i,t}$ (Control vector): Includes regional economic indicators and allows for variation across the time horizon h. (Controls will include variables such as unemployment (STATISTIK AUSTRIA, 2025a) and population growth (STATISTIK AUSTRIA, 2025b).)

The central hypothesis of this thesis is that the coefficient β_h will be statistically significant and positive following monetary easing shocks (e.g. the announcement of the APP/PEPP). This indicates that supply-inelastic districts experience a larger price effect relative to supply-elastic districts.

2.1 Data

- $y_{i,t+h} - y_{i,t-1}$: The cumulative log-change of residential real estate prices in district i from the quarter preceding the monetary shock to h quarters ahead.[2]
- Shock_t: Monetary policy shock derived from the Euro Area Monetary Policy Database (EA-MPD) (Altavilla et al., 2019). This variable captures the exogenous change of various asset prices around ECB policy announcements, isolating the shock components of the APP/PEPP announcements and interest rate decisions.
- Inelastic_i: A dummy-variable that identifies structural district building land supply. This is categorized using the ÖROK dataset on building land reserves (ÖROK Monitoring of Building Land Reserves, 2025).

3 Conclusion and Expected Contributions

The aim of this thesis is to contribute to existing literature on monetary policy transmission by providing a district-level analysis of Austrian residential real estate markets. The central goal of this analysis is to examine how aggregate ECB policy shocks interact with local building land constraints in regional real estate market outcomes. By employing the local projections framework (Jordà, 2005, 2023), this exercise enables an assessment of both time-varying and heterogeneous effects of monetary policy on residential real estate prices (Jordà, 2023). The anticipated result is that supply-inelastic districts exhibit increased and persistent price responses following monetary easing shocks. This expected result is consistent with international findings that housing supply elasticities are crucial in moderating transmission effects through prices (Aastveit and Anundsen, 2018; Albuquerque, 2024).

Potential findings of this analysis may have direct implications for Austrian housing policy. While ECB monetary policy is necessarily uniform, effects on residential real estate markets are not. Evidence for Austria suggests that

[2]This thesis intends to use quarterly district-level residential real estate price data provided by Data Science Service GmbH, which compiles price measures based on transaction data for different residential property categories. At the proposal stage, access to the dataset is pending final confirmation.

regions with constrained land supply experience exacerbated price dynamics during periods of monetary easing (Reiner and Musil, 2023). This, in turn, contributes to persistent affordability issues (Mundt et al., 2024). Formally linking monetary policy shocks, identified using high-frequency data on asset prices from the EA-MPD (Altavilla et al., 2019), to regional real estate outcomes, this thesis assesses potential heterogeneous effects across Austria.

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